APJ Multi-Industry Predictive Maintenance Platform

Technical Overview for Solution Architects

Executive Summary

This is a production-ready, config-driven industrial maintenance intelligence platform that demonstrates both predictive and prescriptive maintenance capabilities across five critical APJ industries. Built on Databricks Lakehouse, it showcases the full spectrum of Data Intelligence Platform capabilities—from OT/IoT ingestion via Zerobus, through medallion architecture with Delta Live Tables, to ML-powered predictions with MLflow, operational decision persistence in Lakebase, and natural language interaction through Genie.

5	20	5
Industries Covered	APJ Sites	Native Currencies

7
Core Technologies

Industry Coverage & Anchor Accounts

Industry	Anchor Account	Use Case Context	Asset Types
Mining	Rio Tinto	Haul fleet and conveyor reliability	Cat 793F trucks, conveyors, crushers
Energy	Alinta Energy	Wind, BESS, transformer availability	Wind turbines, battery systems, transformers
Water	Sydney Water	Pumping and leak-risk operations	Centrifugal pumps, valve systems
Automotive	Toyota	Press and weld line stop-risk reduction	Hydraulic presses, robotic welders
Semiconductor	Renesas	Etch and lithography yield protection	Etch chambers, lithography steppers

APJ Regional Coverage (20 Sites)

Each industry skin includes **four strategically located APJ sites** with site-specific currency and localization, making this demo reusable across the entire APJ region:

Country/Region	Sites per Industry	Native Currency	Language Support
Australia	5 (Rio Pilbara, Sydney Water, Alinta Hornsdale, etc.)	AUD	English

Country/Region	Sites per Industry	Native Currency	Language Support
India	5 (Tata Odisha, Chennai Water, Mahindra Pune, etc.)	INR	English
ASEAN	5 (PETRONAS, PUB Singapore, VinFast, Infineon, etc.)	SGD	English
South Korea	3 (POSCO, Seoul Water, Hyundai Ulsan, KEPCO, Samsung)	KRW	Korean + English
Japan	2 (Toyota Motomachi, Renesas Naka)	JPY	Japanese + English

Databricks Technology Stack

Zerobus

First-mile OT connectivity

Real-time ingestion from industrial protocols (OPC-UA, Modbus, MQTT). Simulator fallback for demo environments. Swinging Door Trending (SDT) compression reduces data volume by 85-95% while preserving signal fidelity.

Delta Live Tables

Declarative ETL pipeline

Three-tier medallion architecture:

- **Bronze:** Raw telemetry normalization
- **Silver:** Feature engineering, OT+PI alignment
- **Gold:** Prediction-ready vectors, financial events

MLflow

ML lifecycle management

Lakebase

Operational OLTP layer

Per-asset anomaly detection (Isolation Forest) and RUL models. Automatic model registration, versioning, and champion/challenger tracking. Centralized experiment tracking at /Shared/ot-pdm-intelligence/{industry}.

Persists operator decisions (Approve/Reject/Defer actions) with sub-second latency. Bridges analytical predictions in Delta Lake with real-time operational systems. Provides durable audit trail for compliance.

Genie

Natural language AI agent
Industry-specific Genie rooms for each vertical. Context-aware queries with currency/language localization. Manual retrieval pattern for equipment documentation. Direct deep-linking from hierarchy maps.

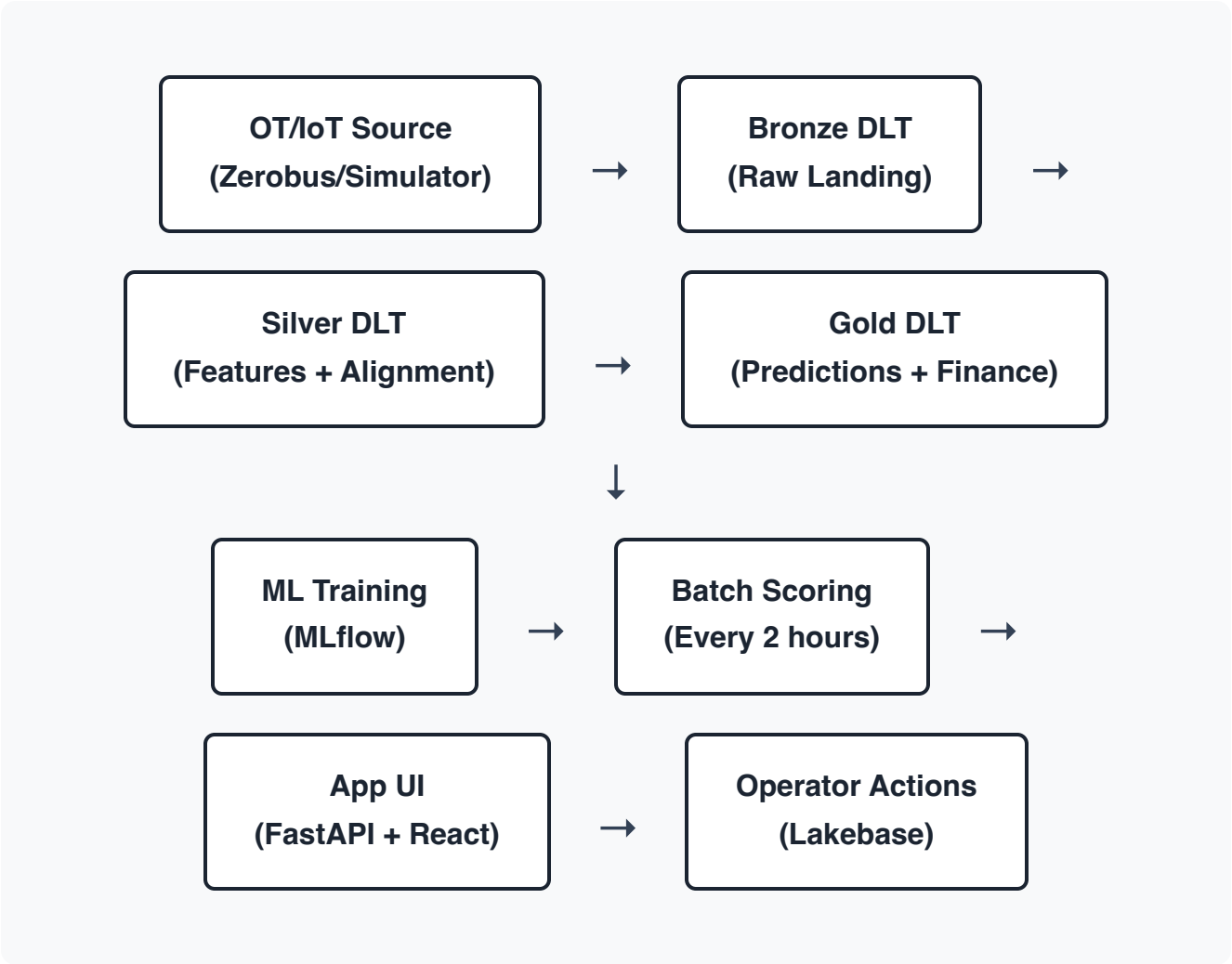
Databricks Apps

Interactive web application
FastAPI backend + React frontend deployed as serverless app. Real-time fleet monitoring, ISA-95 hierarchy navigation, factory map visualization, financial impact dashboards. Currency-aware rendering.

Asset Bundles

Infrastructure-as-Code
Single databricks.yml defines all resources: DLT pipelines, ML training/scoring jobs, Zerobus connector, financial backfill, app deployment. Deploy any industry in under 5 minutes.

Architecture & Data Flow



Key Technical Patterns

1. OT + PI Data Alignment

The platform integrates both OT (Operational Technology) streams and PI (OSIsoft/Aveva) historian data:

- OT rows land to Bronze from Zerobus connector or simulator
- PI rows land to `bronze.pi_simulated_tags`
- Silver creates `ot_pi_aligned` table with time-proximity joins
- Provenance tracking: `BOTH`, `OT_ONLY`, `PI_ONLY`, `UNKNOWN`
- Data source context flows through to UI and downstream analytics

2. Predictive + Prescriptive Intelligence

Unlike pure predictive maintenance tools, this platform closes the execution gap:

- **Predictive:** Anomaly scores, RUL estimates, health degradation trends
- **Prescriptive:** Recommended intervention windows, cost-benefit analysis, expected failure cost vs. planned maintenance cost, actionable work orders
- Gold table `financial_impact_events` merges predictions with operational planning inputs
- Finance dashboard shows avoided costs, ROI projections, and business impact in native or selected currency

3. Currency & Localization

Full multi-currency and multi-language support for APJ deployment:

- **Supported currencies:** USD, AUD, JPY, INR, SGD, KRW, plus AUTO mode (site-native)
- **Language behavior:**
  - JPY → Japanese UI text and Genie prompts
  - KRW → Korean UI text and Genie prompts
  - Other currencies → English with currency conversion
- Backend computes native financial values and converts to selected display currency
- Geo APIs accept currency parameter and return localized financial data
- Executive financial cards render in appropriate currency format with locale-specific number formatting

4. Manual Retrieval Pattern (RAG-lite)

Lightweight grounded retrieval for equipment documentation:

- Manual ingestion from `industries/{skin}/manuals` directory and UI uploads
- Text extraction for markdown, text, and PDF files
- Chunk storage in `bronze.manual_reference_chunks` per industry
- Prompt-time ranking by token overlap and cosine similarity
- Source citations in Genie responses for traceability and trust

5. Swinging Door Trending (SDT) Compression

Production-grade telemetry compression at simulator/connector stage:

- Industry-standard SDT algorithm with configurable epsilon (absolute and percentage)

- Tag-specific overrides for precision-critical sensors (e.g., vibration: 0.06 abs tolerance)
- Heartbeat mechanism ensures maximum gap between transmissions
- Typical compression: 85-95% reduction in row count while preserving anomaly detection accuracy
- Reduces network bandwidth, storage costs, and downstream processing load

UC Business Semantics Layer

Semantic Contract Across Data, APIs, and UX

The platform implements a **Unity Catalog business semantics contract** that standardizes how raw telemetry, model outputs, and finance events are translated into executive metrics. This keeps Genie responses and dashboard KPIs consistent for every industry and site.

Semantic Category	Canonical Meaning	Primary Data Plane	Surface Layer
Risk Semantics	Severity and urgency derived from anomaly + RUL context	Gold `pdm_predictions`	Fleet risk matrix, Geo cards, Genie answers
Value Semantics	Expected failure economics vs intervention economics	Gold `financial_impact_events`	Finance dashboard, recommendation banners
Adoption Semantics	Prediction consumption and action	Finance daily rollups + Lakebase work orders	Executive adoption insights, Genie ROI Q&A

Semantic Category	Canonical Meaning	Primary Data Plane	Surface Layer
	conversion by site		
Governance Semantics	Lineage, ownership, and policy context for metrics	Unity Catalog metadata + audit events	Trust/compliance narrative for CXO and risk teams

- **API semantics:** Finance and executive endpoints expose semantically normalized fields (for example, ROI, utilization, maturity) rather than raw-only counters.
- **Genie semantics:** Prompt context includes adoption/ROI semantics, enabling consistent executive answers without hardcoded responses.
- **Localization semantics:** Currency/language conversion preserves metric meaning while adapting format and language output.

Where UC Metrics Are Defined

- **Primary computation:** `app/server.py` in `_adoption_insights(...)`.
- **Executive payload wiring:** `app/server.py` in `_executive_value(...)`, where `adoption_insights` is attached to the executive response.
- **Genie semantic grounding:** `app/server.py` in `agent_finance_chat(...)`, where adoption metrics are injected into finance chat context.
- **UI rendering:** `app/src/App.jsx` in the executive finance view (Platform Adoption Insights card).

How UC Metrics Are Calculated

Metric	Calculation	Primary Inputs
model_utilization_rate_pct	`(total_actioned_assets / total_alerted_assets) * 100`	Actioned assets (recommendatior

Metric	Calculation	Primary Inputs
		work-order proxy) assets from 30d e
genie_queries_30d	Per-site proxy: `alerted_assets*8 + actioned*12 + total_work_orders*2`, then summed across sites	`gold.financial_i` `lakebase.work_or` actioned asset se
predictions_consumed_sites	Count of sites where `alerted_assets_30d > 0`	Per-site maturity `_adoption_insigh`
cost_per_prediction	`intervention_30d / event_count_30d`	Intervention totals volume from 30d aggregation (with fallback)
avoided_cost_per_prediction	`avoided_30d / event_count_30d`	Avoided-cost tota volume from 30d aggregation (with fallback)
platform_roi_x	`ebit_saved / (intervention_cost + platform_cost)`	Executive EBIT s intervention/platfo investment from e value computatio
mttr_improvement_pct	Proxy curve: `min(35, 6 + model_utilization_rate_pct*0.12)`	Global model utili
failure_recurrence_reduction_pct	Proxy curve: `min(35, 8 + model_utilization_rate_pct*0.15)`	Global model utili

Metric	Calculation	Primary Inputs
site_maturity.maturity_score	`0.55*action_rate + 0.25*prediction_coverage + 0.20*engagement_score`, bounded to 0-100	Action rate, predi coverage, and no engagement (`genie_queries_3`

Quick Deployment Guide

Deploy a New Industry in Under 5 Minutes

Step 1: Validate Bundle

```
# Validate configuration before deployment databricks bundle
validate --target dev -p DEFAULT
```

Step 2: Deploy Infrastructure

```
# Deploy specific industry (mining, energy, water, automotive,
semiconductor) databricks bundle deploy --target dev -p DEFAULT -
-var industry=mining # Or deploy all industries databricks bundle
deploy --target dev -p DEFAULT
```

Step 3: Bootstrap Data & Jobs

```
# Run the bootstrap notebook in Databricks workspace:
RUNME_BOOTSTRAP_ALL.py # This provisions all configured
industries: # - Catalog and schema creation # - Core DDL (tables,
views) # - Seed data and planning cases # - Finance history
backfill (730 days default) # - Permissions grants # - Optional:
trigger ML training and scoring jobs
```

Step 4: Verify & Launch

```
# Verify data landed in Bronze/Silver/Gold tables SELECT * FROM
pdm_mining.bronze.sensor_readings LIMIT 10; # Open app endpoint
from Databricks Apps console
https://<workspace>.cloud.databricks.com/serving-endpoints/ot-
pdm-app
```

Configuration-Driven Reusability

The platform's config-driven architecture makes it **trivially reusable for SA teams across APJ**:

- **Industry skins:** Each industry has a `config.yaml` defining assets, hierarchy, anchor accounts, and simulator parameters
- **Site expansion:** Add new sites by extending the sites array with site_id, name, country, currency
- **Asset templates:** Define asset types once, instantiate across sites with fault profiles
- **Deployment matrix:** `industries/deployment_matrix.yaml` enforces resource consistency (jobs, pipelines) across all verticals
- **Reconciliation tooling:** `tools/reconcile_industry_matrix.py` audits and auto-corrects resource drift

Demo Scenarios for Customer Meetings

Scenario 1: Real-Time Fleet Monitoring (Operations Persona)

- Show live asset health across 20 APJ sites with Geo hierarchy navigation
- Drill down to specific asset telemetry and anomaly scores
- Demonstrate factory map mode with physical layout and live status pins
- Click "Investigate in Genie" to handoff asset context to AI agent

- Switch currency selector to show localized financial impact (e.g., INR for India sites)

Scenario 2: Predictive Maintenance with ROI (Manager Persona)

- Navigate to Model page to show anomaly detection and RUL predictions
- Highlight specific assets approaching failure threshold
- Show financial impact: expected failure cost vs. planned intervention cost
- Demonstrate avoided cost calculations (prescriptive value)
- Generate work order recommendation with intervention window

Scenario 3: Natural Language Analytics (Executive Persona)

- Open Genie room for the active industry
- Ask: "What are the top 3 highest-risk assets across all sites?"
- Ask: "Show me total avoided costs for the last 30 days in AUD"
- Switch to JPY currency and observe Japanese-localized Genie responses
- Upload equipment manual PDF and ask: "What is the recommended bearing replacement procedure?"

Scenario 4: OT Connectivity & Data Pipeline (IT/Architect Persona)

- Show Zerobus connector configuration and protocol support
- Demonstrate SDT compression metrics (85-95% reduction)
- Navigate through DLT pipeline DAG (Bronze → Silver → Gold)
- Show OT+PI alignment table and data provenance tracking
- Highlight MLflow experiment tracking and model versioning
- Show Lakebase action persistence and audit trail

Reusability Across APJ Field Teams

Why This Demo is Perfect for APJ SAs

- **Multi-industry coverage:** One codebase covers 5 major verticals—mining, energy, water, automotive, semiconductor. No need to maintain separate demos.
- **Geo-distributed sites:** 20 pre-configured APJ sites across Australia, India, ASEAN, Korea, Japan. Demo using local anchor accounts for credibility.
- **Currency & language localization:** Native support for AUD, INR, SGD, KRW, JPY with automatic UI and Genie localization. Works out-of-the-box for local execs.
- **Fast deployment:** Deploy new industry in under 5 minutes with Databricks Asset Bundles. No manual infrastructure setup.
- **SFDC-aligned:** Anchor accounts match real APJ opportunities (Rio Tinto, Alinta, Sydney Water, Toyota, Renesas). Pipeline context is built-in.
- **Full stack demonstration:** Shows complete Databricks platform: Zerobus, DLT, MLflow, Lakebase, Genie, Databricks Apps. Not just a partial POC.
- **Production-grade patterns:** SDT compression, OT+PI alignment, medallion architecture, model lifecycle management, operational decision persistence—all enterprise-ready.

Technical Differentiation

Capability	Traditional PdM Solutions	This Platform
Data Integration	Manual ETL, limited protocol support	Zerobus native OT connectivity (OPC-UA, Modbus, MQTT) with SDT compression

Capability	Traditional PdM Solutions	This Platform
Architecture	Proprietary data silos, rigid schemas	Open Delta Lake medallion, config-driven extensibility
Intelligence	Predictive only (what will fail)	Predictive + Prescriptive (what to do, when, at what cost)
ML Lifecycle	Manual model management, version chaos	MLflow automatic versioning, champion/challenger, A/B testing
Operational Integration	Read-only analytics, no feedback loop	Lakebase for operator decisions, Zerobus writeback, full closed-loop
User Experience	SQL dashboards for analysts only	Genie natural language for executives, interactive React app for ops
Global Deployment	English-only, single currency	Multi-language, multi-currency, site-aware localization

Resources & Next Steps

Documentation

Complete README with architecture diagrams, deployment guides, and API documentation available in repo root.

Video Walkthrough

Record a 10-minute Loom showing end-to-end flow from deployment to Genie interaction for SA enablement.

Customization Guide

Step-by-step instructions for adding new industries, sites, asset types, and

Support Channel

APJ Field Engineering Slack channel for questions, demo requests, and

custom financial models.

feature suggestions.

Ready to Deploy?

Start your demo environment in under 10 minutes:

1. Clone the repo to Databricks workspace
2. Run `databricks bundle deploy --var industry=mining`
3. Execute `RUNME_BOOTSTRAP_ALL.py` notebook
4. Open the app from Databricks Apps console
5. Schedule your first customer demo!

Questions? Reach out to the APJ Field Engineering leadership team or consult the deployment matrix reconciliation guide for troubleshooting.